EcoEnergy Insight

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Hypothesis

We hypothesize that there are significant changes and relationships in agricultural total factor productivity (TFP), energy consumption, and carbon dioxide emissions across various countries from 1993 to 2021. Our motivation is to explore these dynamics to inform sustainable agricultural practices and energy policies, considering the growing concerns about climate change and resource depletion.

Data

The data comprises three main datasets: 1) Agricultural indices (TFP, capital, labor, land) from 1993 to 2021 sourced from the USDA; 2) Energy consumption and CO2 emissions data from OurWorldInData; and 3) Additional environmental and economic metrics. The datasets were cleaned to exclude years with incomplete data and focused on modern periods for relevance and accuracy.

Findings

Claim #1: There was a significant change in agricultural TFP by countries in the past decade.
Support for Claim #1: Utilizing a Paired T-Test, we found a significant statistical change in the TFP production in countries from 2010 to 2021, with a **p-value of 0.00037**. This suggests that agricultural productivity has notably evolved during this period.

Claim #2: Between the United States and China, there was a statistically significant difference in energy consumption per capita from 1993 to 2021.
Support for Claim #2: A Two Sample T-Test revealed a substantial difference in average energy per capita between the US and China, with a **p-value of 6.97937e-51**, indicating divergent energy utilization patterns between these two major economies.

Claim #3: There was a significant change in cumulative carbon dioxide emissions for countries from 2018 to 2021.
Support for Claim #3: Another Paired T-Test showed significant changes in cumulative CO2 emissions among countries during this short period, with a **p-value of 0.0099432**, highlighting rapid changes in emission levels possibly influenced by policy shifts or economic changes pre-pandemic.

Sociohistorical Context:

The intersection of climate change, energy consumption, and agricultural productivity is influenced by a range of socio-historical factors, including governmental policies on environmental regulation, global events like the Paris Agreement, and the technological

advancements in renewable energy. These factors collectively shape how nations manage their agricultural sectors amidst climate challenges. For instance, stricter CO2 emission standards in many countries may lead to shifts in energy consumption patterns, influencing both our data and our hypothesis about the relationship between energy use and agricultural productivity.

Stakeholders

Farmers: these are the primary producers cultivating the land who are affected by various factors such as climate change, soil richness, land availability, energy sources, etc… Changes in agricultural practices are well associated with energy use and climate change

Global Consumers: grocery stores purchase the available supply of marketable goods from farmers and of course these are then put on the market for people to consume/purchase.As end-users of agricultural products, changes in production efficiency and sustainability can affect food security and prices.

Government and Energy Companies: the analysis of this data could help bring more regulations to how energy companies expand as well as how much we consume. It can also guide us into a direction of implementing more alternative energy sources and especially those that support farmers such as solar panels, water energy nexuses, and more.

Animals/Plants: Society heavily relies on the use of breeding and NGOs in order to massively increase market supply. Carbon dioxide quantities also affect how their habitats change over time. Changes revolving around this study could reduce our reliance on the use of these chemicals and fossil fuels which harm the environment they inhabit and how we treat them. Interested in promoting sustainable practices and might use the research findings to advocate for policy changes. These stakeholders are interlinked.

Impact

Previous studies have often focused on the micro-level impacts of climate change on local farming practices without integrating the broader energy consumption perspective. Research showing a correlation between increased energy efficiency and improved agricultural outcomes has led to policy shifts towards more sustainable farming subsidies and energy practices. The analysis should consider historical emissions data and current energy policies to understand long-term trends. **Interpretations of findings should be contextualized within the broader socio-political environment, acknowledging that while some countries contribute more to global emissions, others suffer disproportionately from the impacts of climate change.**

Some biases that may exist in the interpretation of our data is our inherent perception of how technologically advanced and industrialized some countries are in comparison to others. Additionally, for some countries that aren't as industrialized, we may see them as more suited for high agricultural productivity, specifically regions in Latin America and Caribbean and Africa. On the other hand, we inherently believed that regions such as the United States and China would be astronomical outliers in the data due to them being superpower countries, so they may be high in agricultural productivity, CO2 emissions, and the other variables captured within our datasets.

In our decision making process across the lifetime of the project, we decided to filter out years where comprehensive data was not captured. In other words, for all the variables captured within our datasets, we decided to completely drop rows of data where there may have been a null value

contained within the dataset. Although this may have brought us to a more comprehensive conclusion, it may have affected us when deciding to perform a regression analysis for two reasons. One, because of the null values (especially in data that was captured in the early 1900s), we decided to look between the specific years of 1993-2021. This decision also influenced us to perform a regression on the totality of all the variables in the datasets. In other words, for some columns we decided to average the values, and for a column Total_Outall, we decided to total all the values for the given years they were provided. In hindsight, the decision made sense. However, if we were to revisit, instead of performing a regression on the amalgamation of the agricultural TFP index and cumulative CO2, we would heavily consider plotting the data for every year for each country so we can potentially have a more accurate prediction of the relationship between TFP and carbon dioxide emissions. As a result, some interpretations of the regression plot may be seen as not valuable due to the outliers presented by the visualization. As a whole, our collective identities, prior knowledge, and perspective extensively informed our beliefs when working on this project. When building our hypotheses, we were well aware of needing to choose a time period, where generally, we could expect a majority of countries to be well-industrialized. Of course this is not representative of the actual state of the world itself, but the contemporary time frame allowed us to generally gauge our expectations. Additionally, we knew that superpower countries such as the United States and China, and countries who have extreme agricultural value may be high in certain indices and low in others. Furthermore, countries have extremely different cultures, perspectives, geographies, and other factors that may extremely influence how the data is interpreted and the symbolization of their values.

Ethical Considerations

Some regions, as stated before, may have a skewed contribution towards climate change as well as being unfairly impacted by the effects of climate change. It is important to understand the historical biases that may form as a result of the data gathered throughout the project and inform ourselves about these confounding factors. Additionally, our findings suggesting certain regions are more vulnerable to climate impacts could be misinterpreted as those regions being less suitable for investment. Proactively clarifying the implications and advocating for supportive policies rather than punitive measures can mitigate such risks. Lastly, we have communicated the limitations and scope of the project to help convey and support its ethical integrity. As college students, the study is designed to highlight and address global disparities in the impact of climate change, aiming to contribute to more equitable policy interventions. We intentionally designed our hypotheses to cover specific areas:

- A decade analysis found within our first hypothesis that implicitly attempts to measure the effect of technology advancements within a decade on agriculture productivity. We wanted to stay within the timeframe as it is a more contemporary time period, as well as we didn't want to have such a large interval of time where there was an extremely stark contrast in things such as culture, technological advancements, perception of climate change, and more. Although all of the aforementioned are essential to consider, it may be oblivious to make the statement that "current day technology enables agriculture productivity to be significantly higher than it was compared to a 1932"
- An analysis of the energy usage of countries that have very large populations and a vast amount of resources to consider their energy consumption. In addition, they are some of the largest emitters of carbon dioxide and other greenhouse gasses. It is important to consider how nonrenewable energy initiatives go forward with this context in mind.

- The effect of the pandemic on carbon dioxide emissions and how they compare to the years leading up to the pandemic. It is important to statistically understand the effects in which the shock of the coronavirus pandemic may impacted the economy and policies that may have been put forth, in addition to how those two facets may have affected emissions, if at all.